

MATH 411: Week 8

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Lemma 1

Suppose f is defined on an interval (a, b) with $f(x) < M$ for all $x \in (a, b)$, and let $p \in (a, b)$. Then,

$$L = \limsup_{x \rightarrow p^+} f(x) \leq M.$$

A similar result holds for minima and limits inferior.

Since f is bounded on (a, b) , we already know L is finite. Suppose $L > M$ and let $\epsilon = L - M > 0$. Then if there's an $x \in (p, b) \subseteq (a, b)$ such that $f(x) \in (L - \epsilon, L + \epsilon)$, we have

$$f(x) > L - \epsilon = L - (L - M) = M.$$

Since we know $f(x) < M$, no such x exists. We've shown that there's a neighborhood around L containing no outputs of f corresponding to inputs to the right of p . But L is a limit superior, so we conclude $L \leq M$. $\square$

Setup

For Problems 8.21–24, assume that

Let f and g be real, differentiable functions on some open interval I with $a \in I$, and $g'(x) \neq 0$ for all $x \in I$. Suppose

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = A.$$

Also suppose $\lim_{x \rightarrow a^+} f(x) = 0$ and $\lim_{x \rightarrow a^+} g(x) = 0$.

Problem 8.21

Let p and q be any real numbers such that $p < A < q$, and choose an r and s such that $A < r < q$ and $p < s < A$. Let $b \in I$ with

$a < b$. Explain why there exists some $c \in (a, b)$ such that for all x with $a < x < c$ we have

$$s < \frac{f'(x)}{g'(x)} < r.$$

Let $\epsilon = \min\{r - A, A - s\}$, which is positive since $r > A$ and $A > s$. Then there's a $\delta > 0$ such that if $0 < |x - a| < \delta$, then

$$\begin{aligned} \left| \frac{f'(x)}{g'(x)} - A \right| &< \epsilon \\ -\epsilon &< \frac{f'(x)}{g'(x)} - A < \epsilon \\ A - \epsilon &< \frac{f'(x)}{g'(x)} < A + \epsilon, \end{aligned}$$

which we can rewrite:

$$s = A - (A - s) \leq A - \epsilon < \frac{f'(x)}{g'(x)} < A + \epsilon \leq A + (r - A) = r.$$

Let $\gamma = \min\{b - a, \delta\}$ and let $c \in (a, a + \gamma)$. γ is positive, since $b > a$ and $\delta > 0$, so c exists. Further, $c \in (a, b)$, since

$$c < a + \gamma \leq a + (b - a) = b.$$

If $x \in (a, c)$, then

$$\begin{aligned} a &< x < c < a + \gamma \leq a + \delta \\ 0 &< x - a < \delta \\ 0 &< |x - a| < \delta, \end{aligned}$$

and thus

$$s < \frac{f'(x)}{g'(x)} < r.$$

So we've found a c with the desired properties. $\square$

Problem 8.22

Let $x, y \in (a, c)$ with $x < y$. Explain why there exists $t \in (x, y)$ such that

$$s < \frac{f(y) - f(x)}{g(y) - g(x)} = \frac{f'(t)}{g'(t)} < r.$$

Since I is an open interval and both x and y are in I , neither x nor y are the endpoints of I , which ensures f and g are defined and differentiable (and thus continuous) on $[x, y]$. So the Cauchy mean value theorem ensures there's a $t \in (x, y)$ such that

$$(f(y) - f(x))g'(t) = (g(y) - g(x))f'(t).$$

If $g(y) - g(x) = 0$, then the mean value theorem implies that there's a $p \in (x, y)$ such that

$$\begin{aligned}(y - x)g'(p) &= 0 \\ g'(p) &= 0,\end{aligned}$$

where the last equality is justified since $x \neq y$. However, by assumption $g'(p) \neq 0$, so we conclude $g(y) - g(x) \neq 0$. Also, $g'(t) \neq 0$. Now we can write

$$\frac{f(y) - f(x)}{g(y) - g(x)} = \frac{f'(t)}{g'(t)},$$

as desired. $\square$

Problem 8.23

Prove that for all $y \in (a, c)$,

$$p < s \leq \frac{f(y)}{g(y)} \leq r < q.$$

Let $y \in (a, c)$, and consider

$$\begin{aligned}\lim_{x \rightarrow a^+} \frac{f(y) - f(x)}{g(y) - g(x)} &= \frac{\lim_{x \rightarrow a^+} (f(y) - f(x))}{\lim_{x \rightarrow a^+} (g(y) - g(x))} \\ &= \frac{f(y) - \lim_{x \rightarrow a^+} f(x)}{g(y) - \lim_{x \rightarrow a^+} g(x)} \\ &= \frac{f(y)}{g(y)}.\end{aligned}$$

Let $\alpha \in I$ with $\alpha < a$ and let $\beta \in (y, c)$. α exists, since I is an open interval and $a \in I$. Since $(\alpha, \beta) \subseteq I$, both f and g are defined on (α, β) . The previous problem ensures that for any $x \in (\alpha, \beta)$, we have

$$s < \frac{f(y) - f(x)}{g(y) - g(x)} < r.$$

Lemma 1 now ensures

$$s \leq \liminf_{x \rightarrow a^+} \frac{f(y) - f(x)}{g(y) - g(x)} = \lim_{x \rightarrow a^+} \frac{f(y) - f(x)}{g(y) - g(x)} = \limsup_{x \rightarrow a^+} \frac{f(y) - f(x)}{g(y) - g(x)} \leq r.$$

$\square$

Problem 8.24

Conclude the proof of the theorem, that is that

$$\frac{f(x)}{g(x)} \rightarrow A \text{ as } x \rightarrow a \text{ (from the right).}$$

The previous problem ensures that

$$p < \frac{f(y)}{g(y)} < q.$$

So Lemma 1 ensures

$$p \leq \liminf_{x \rightarrow a^+} \frac{f(y)}{g(y)} \leq \limsup_{x \rightarrow a^+} \frac{f(y)}{g(y)} \leq q.$$

Since p and q are arbitrary real numbers with $p < A < q$, we conclude that

$$\liminf_{x \rightarrow a^+} \frac{f(y)}{g(y)} = A = \limsup_{x \rightarrow a^+} \frac{f(y)}{g(y)}.$$

□